

32. Joint with R. Brauer: On minimal splitting fields of irreducible representations

Sitz. Ber. d. Preuß. Akad. d. Wiss. 1927, pp. 221–228.

(Presented by Mr. Schur.)

The question of the number fields of smallest degree over a ground field P in which a representation of a finite group that is irreducible over P decomposes into absolutely irreducible constituents was first treated by I. Schur.¹ Suppose, for example, that P contains the character of such a constituent. Schur then showed that the degree of these fields relative to P – the index – agrees with the number of those absolutely irreducible constituents which all belong to the same class; that consequently the same field is already determined by the requirement of splitting off one absolutely irreducible constituent; and further that the degree of every field, relative to P , in which such a splitting occurs is a multiple of the index. All these fields are to be called splitting fields, also in the case where P does not contain the character. By an example Schur showed that the splitting fields of smallest degree need not be isomorphic. If P contains no corresponding simple character, then the splitting fields must comprise the field of such a character and its conjugates with respect to P ; the absolutely irreducible representations belonging to different conjugate characters are indeed conjugate, but plainly belong to different classes. To split off a system of absolutely irreducible representations of one class, here too it suffices to take the field of the corresponding character and one of the corresponding splitting fields.

In what follows the question of splitting fields is pursued further. The splitting fields of smallest degree can be characterized by assigned non-commutative division rings; the general splitting fields by two-sided simple rings invariantly connected with these non-commutative division rings. Further, by means of the example of the quaternion division ring, it is shown that the degrees of minimal splitting fields are not bounded; here a splitting field is called minimal if none of its proper subfields is a splitting field. This unboundedness follows essentially from a number-theoretic existence theorem whose proof is supplied by H. Hasse in the immediately following note. The example, however, is also treated elementarily by computation – more in the way of verification; and thus, without using the general theory and the number-theoretic existence theorem, the unboundedness of the degrees is shown by proving elementarily a less far-reaching, but sufficient, existence theorem.² It should also be noted that in this example we are again dealing with splitting fields of a finite group, namely the quaternion group, which also underlies Schur's example. The representations given there are simultaneously (isomorphic) representations of the quaternion division ring.

1. Characterization of the splitting fields

First let us briefly assemble the basic concepts. Let $\mathfrak{S}$ denote a hypercomplex system with respect to a field P_0 . By a representation Γ of $\mathfrak{S}$ – in P – one means a system of matrices with elements from P , in such a way that Γ becomes a homomorphic image of $\mathfrak{S}$, and that diagonal matrices $p_0 E$ are assigned to the elements p_0 from P_0 ; hence P must contain P_0 . A representation Γ together with all its transforms $P^{-1}\Gamma P$ forms a representation class. The representations of finite groups G are included in the representations so defined; the assigned hypercomplex system $\mathfrak{S}$, the “group ring”, consists of all “group numbers”, i.e. of all linear combinations of the group elements, with coefficients from P_0 , the original group elements being regarded as linearly independent and multiplication being defined by group multiplication and the laws of arithmetic. The notions “reducible”, “irreducible”, “completely reducible”, and “absolutely irreducible” representations are

¹Arithmetische Untersuchungen über endliche Gruppen, Sitzungsber. d. Berl. Ak. d. Wiss. 1906, p. 164. Beiträge zur Theorie der Gruppen linearer Substitutionen, Transact. of the Am. Math. Soc. Ser. 2, Vol. XV, 1909, p. 159. In the second paper the results are transferred to completely reducible hypercomplex systems. A representation is collected into one class together with all its transforms (similar representations).

²This example is due to E. Noether; it rests on a modification of one due to R. Brauer, in which the existence of a minimal splitting field whose degree exceeds the index was shown in general. The elementary treatment of the example is due to R. Brauer. The characterization of the splitting fields goes back to investigations of the authors, which otherwise pursue separate goals. For E. Noether the matter is a construction of representation theory on the basis of module and ideal theory, under general finiteness hypotheses; for R. Brauer it is the construction of non-commutative division rings of finite degree over a given commutative perfect ground field, with the help of factor systems. The authors arrived independently at the characterization of splitting fields of smallest degree, R. Brauer for perfect ground fields, E. Noether without this restriction. R. Brauer found the characterization of the general splitting fields in connection with a question of E. Noether concerning the possibility of non-commutative embedding; E. Noether was able here also to remove the restriction to a perfect ground domain.

defined as usual; the terminology is to be extended to representation classes. A system $\mathfrak{S}$ is called “completely reducible in P ” if all its representation classes in P are completely reducible.

Now assume $\mathfrak{S}$ to be completely reducible in P_0 ; assume P_0 to be a perfect field, so that the same is true for every algebraic extension P of P_0 . The representation classes remain completely reducible under every extension P of P_0 .³ The center of $\mathfrak{S}$ becomes a direct sum of finitely many fields; every extension P of P_0 isomorphic to such a field K is the field of a (simple) character. If P_0 is extended to a P isomorphic to K , then there splits off from $\mathfrak{S}$ a two-sided simple ring to which the absolutely irreducible representation class of the character is assigned. By Wedderburn’s theorem this ring is the direct product of a matrix ring over P with a non-commutative division ring $\mathfrak{A}$, uniquely determined up to isomorphism, whose center is P and which has degree m^2 relative to P , where m denotes the Schur index belonging to the character. Every splitting field of $\mathfrak{S}$ – with respect to the representation belonging to the character – is at the same time one of $\mathfrak{A}$ with respect to the representations in P , and conversely. The conjugate representations of $\mathfrak{S}$ are obtained by starting from the corresponding two-sided simple ring over P_0 ; its assigned division ring $\mathfrak{A}_0$ becomes isomorphic to $\mathfrak{A}$ and has K as center. Thus the problem of splitting fields is completely reduced to that of the assigned non-commutative division ring $\mathfrak{A}$ and its splittings.⁴ In particular one has the theorem:

All maximal commutative subfields of $\mathfrak{A}$ have the same degree m relative to P , where m denotes the Schur index. All these maximal commutative subfields are splitting fields of smallest degree, and every splitting field of smallest degree is contained among them. The degree of every splitting field is a multiple of m . The ring $\mathfrak{A}$ has only one absolutely irreducible representation class and likewise only one irreducible representation class with respect to P .

For the characterization of the general splitting fields, let $\mathfrak{A}_r$ denote the direct product of $\mathfrak{A}$ with the matrix ring of degree r^2 over P (the system of all r -rowed matrices with elements from P). Then $\mathfrak{A}_r$ is two-sided simple, with $\mathfrak{A}$ as assigned non-commutative division ring; and every two-sided simple ring, hypercomplex over P , to which $\mathfrak{A}$ is assigned, is obtained in this way. $\mathfrak{A}_r$ is also characterized as the system of all r -rowed matrices with elements from $\mathfrak{A}$.

Every splitting field of $\mathfrak{A}$ of degree mr – if such a field exists – is isomorphic to a maximal commutative subfield of $\mathfrak{A}_r$. Conversely, all maximal commutative subfields of $\mathfrak{A}_r$ are splitting fields, each of degree ms , where s is a divisor of r . In the regular case all maximal commutative subfields of $\mathfrak{A}_r$ have degree mr . Here by the regular case is meant: the ground field P has the property that for every commutative field Σ over P which is finite over P , there are still commutative extensions over Σ of arbitrary degree.⁵

Whether among these splitting fields, for $r > 1$, minimal ones also occur is not yet decided by the embedding in $\mathfrak{A}_r$. That this actually can be the case for infinitely many r is shown by the following example.

2. The unboundedness of the degrees of minimal splitting fields

For this unboundedness the quaternion division ring is now to be given as an example, regarded as a hypercomplex system over the field P of rational numbers, with the basis elements i, j, k besides the unit 1 and with the familiar relations:

$$i^2 = j^2 = k^2 = -1; \quad ij = k; \quad jk = i; \quad ki = j; \quad ji = -k; \quad kj = -i; \quad ik = -j.$$

It will be shown that all and only those algebraic number fields are splitting fields for which, after adjunction, there exists an “idempotent” element r , so that $r^2 = r$, with r different from both the zero element and the unit element. These number fields can also be characterized as those in which (-1) is representable as a sum of three squares, and consequently as a sum of two squares.⁶ Indeed, set

$$r = \alpha + \frac{1}{2}\beta i + \frac{1}{2}\gamma j + \frac{1}{2}\delta k;$$

³If P_0 is not perfect, complete reducibility is preserved if and only if the components K of the center become “extensions of the first kind” of P_0 . Under this hypothesis all further consequences in the text remain valid.

⁴This corresponds to Schur’s reduction to the system of matrices over P which commute with the irreducible representation of $\mathfrak{S}$ in P . The transposes of these matrices give a representation of $\mathfrak{A}$ in P .

⁵Thus, for example, no regular case occurs when P is the field of all real numbers.

⁶That representability of (-1) as a sum of three squares always implies such representability as a sum of two follows from the identity

$$(c^2 + d^2)(a^2 + b^2 + c^2 + d^2) = (ac + bd)^2 + (ad - bc)^2 + (c^2 + d^2)^2,$$

which follows, for example, from the product theorem for norms of quaternions. Instead of requiring the existence of an idempotent element, it would also have sufficed for the splitting to require an element of vanishing norm.

then

$$r^2 = \left(\alpha^2 - \frac{1}{4}\beta^2 - \frac{1}{4}\gamma^2 - \frac{1}{4}\delta^2 \right) + \alpha\beta i + \alpha\gamma j + \alpha\delta k,$$

so that $r^2 = r$ is equivalent to

$$4\alpha^2 - 4\alpha - \beta^2 - \gamma^2 - \delta^2 = 0; \quad 2\alpha\beta - \beta = 0; \quad 2\alpha\gamma - \gamma = 0; \quad 2\alpha\delta - \delta = 0,$$

and hence, since β, γ, δ are not to vanish simultaneously, equivalent to

$$(-1) = \beta^2 + \gamma^2 + \delta^2.$$

That every such idempotent element produces a splitting, and that the existence of such an idempotent element is necessary for the splitting, follows from these facts: every irreducible representation of a completely reducible system is generated by a one-sided simple ideal, more precisely by the corresponding ideal class, and every such ideal class generates the representation. These one-sided simple ideals all occur in a one-sided direct-sum decomposition, and this decomposition determines the “primitive” idempotent elements that become the components of the unit. Conversely, every idempotent element gives a direct-sum decomposition $\mathfrak{S} = r\mathfrak{S} + (e - r)\mathfrak{S}$, where e denotes the unit; the “primitive” ones split off simple ideals.⁷ A non-commutative division ring, regarded as a hypercomplex system with respect to its center, becomes – when extended to a hypercomplex system with respect to a splitting field – a direct sum of m absolutely simple one-sided ideals, where m is the Schur index.

In the case of the quaternion division ring, m is two; hence every idempotent element $\neq 0$ and $\neq 1$ already produces the decomposition into absolutely simple ideals to which the splitting into absolutely irreducible representation classes corresponds. This characterizes the indicated fields as splitting fields.

It follows further: all cyclic fields Ω_n of degree 2^n over the field of rational numbers in which (-1) can be represented as a sum of two squares are minimal splitting fields of the quaternion division ring. For a splitting field cannot be real, because of the representation of (-1) as a sum of squares; on the other hand, the subfield of Ω_n of degree 2^{n-1} – and therefore every proper subfield of Ω_n – is necessarily real, being the intersection of Ω_n with the field of all real algebraic numbers. Namely, Ω_n is a Galois field of degree two over this intersection, which therefore must coincide with the subfield of degree 2^{n-1} .

According to Hasse’s existence theorem⁸ there are such fields Ω_n for every n ; the degrees of minimal splitting fields are not bounded. In fact, every power of the index occurs here as the degree of a minimal splitting field.⁹

3. Elementary treatment of the splitting fields of the quaternion division ring

Let $\mathfrak{D}$ denote the representation of the quaternion group formed from the following eight matrices:¹⁰

$$\pm E = \pm \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \pm I = \pm \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \pm J = \pm \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \pm K = \pm \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

⁷The connection between primitive idempotent elements and irreducible representations is already found in the dissertation of M. Herzberger: *Über Systeme hyperkomplexer Größen*, Chapter 4, Berlin 1923. That simple ideals generate irreducible representations is shown in the commutative case, for example, by E. Noether, *Der Diskriminantensatz für Ordnungen*, Crelle 157 (1926); in the non-commutative case by W. Krull, *Theorie und Anwendung der verallgemeinerten Abelschen Gruppen*, Heidelberger Berichte 1926. Cf. also the closing remarks in Speiser’s group theory. If $a_1, \dots, a_m$ denotes a basis of the ideal with respect to the splitting field, and c an arbitrary ring element, so that $a_i c = \gamma_{i1} a_1 + \dots + \gamma_{im} a_m$, then (γ_{ik}) is the matrix assigned to c ; the fact that all irreducible representations are generated by ideals lies deeper.

⁸The existence of Ω_n can already be read from the parameter representation of cyclic fields of degree 4, as given for example in Weber (*Kleines Lehrbuch der Algebra*, § 93).

⁹Subsequently R. Brauer was able to show further that there are even minimal splitting fields of the quaternion division ring of every even degree, hence of all degrees possible at all for splitting fields. Such a field $P(\Theta)$ may be defined for every $n \geq 2$ by the equation

$$f(\Theta) = \Theta^{2n} + a^2 p^2 \Theta^2 + b^2 p^2 \beta^2 = 0; \quad \text{therefore} \quad -1 = \left(\frac{ap}{\Theta^{n-1}} \right)^2 + \left(\frac{bp\beta}{\Theta^n} \right)^2,$$

where, by suitable choice of a, b, p, β , one can achieve that 1. $f(x)$ is irreducible for every n ; 2. $P(\Theta)$ has only the one subfield $P(\Theta^2)$ different from P ; 3. $P(\Theta^2)$ also has real fields among its conjugates, and thus cannot be a splitting field. One possible choice of a, b, p, β is, for example,

$$f(x) = x^{2n} + (2^5 \cdot 9)4x^2 + 9 \cdot 64.$$

The proof rests on a modification of a method of Furtwängler for constructing primitive equations (Math. Ann. 85, p. 34, § 3), but is rather laborious in the detailed execution.

¹⁰Cf. Sylvester, Math. Papers Vol. III p. 647, Vol. IV p. 122.

The group ring generated by $\mathfrak{D}$ is precisely a representation of the quaternion division ring regarded as a hypercomplex system; we shall show elementarily that $\mathfrak{D}$ is representable in a field K if and only if -1 is representable in K as a sum of two squares.

a) Let $\mathfrak{D}^* = T^{-1}\mathfrak{D}T$ be a representation rational over a field K , and put $I^* = T^{-1}IT$, $J^* = T^{-1}JT$. Since J and J^* are two similar matrices rational over K , there is also a transformation R rational over K which carries J^* into J ,

$$R^{-1}J^*R = J.$$

If $\mathfrak{D}^*$ is replaced from the start by the likewise K -rational representation $R^{-1}\mathfrak{D}^*R$, one sees that, without restriction, one may assume $J^* = J$. Let

$$I^* = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad (a, b, c, d \text{ numbers from } K).$$

From $IJ = -JI$ it follows that $I^*J^* = -J^*I^*$, and because $J^* = J$ this gives

$$a = -d, \quad b = c.$$

From $I^2 = -E$ it follows that $I^{*2} = -E$, hence

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}^2 = \begin{pmatrix} a^2 + b^2 & 0 \\ 0 & a^2 + b^2 \end{pmatrix},$$

so $a^2 + b^2 = -1$; consequently -1 is indeed representable in K as a sum of two squares.

b) Conversely, suppose that -1 is representable in a field K as a sum of two squares. If, say, $-1 = a^2 + b^2$, set

$$I^* = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}, \quad J^* = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad K^* = I^*J^*.$$

A straightforward calculation shows that $\pm E, \pm I^*, \pm J^*, \pm K^*$ form a group rational over K and similar to $\mathfrak{D}$.¹¹

It remains to show that for infinitely many n there are cyclic fields of degree 2^n which are minimal splitting fields of the quaternion group.

Let p be a rational prime such that, for some suitable $r > 0$,

$$(*) \quad 2^r + 1 \equiv 0 \pmod{p}$$

and let 2^n be the highest power of 2 dividing $p - 1$.

First we show that primes p occur for infinitely many values of n . Let k be any positive rational integer, and let p be a prime divisor of $2^{2^k} + 1$. Then the congruence required for p is fulfilled with $r = 2^k$. Further, $2 \pmod{p}$ has exponent 2^{k+1} , and therefore, for the highest power 2^n dividing $p - 1$, the number n is greater than k . Thus in any case there are arbitrarily large n to which primes p correspond.

Let ε be a primitive p -th root of unity. We now show that in $P(\varepsilon)$ the number -1 can be represented as a sum of two squares,

$$-1 = a^2 + b^2 = (a + ib)(a - ib) \quad (a, b \text{ numbers from } P(\varepsilon)).$$

This is equivalent to saying that in the field of $(4p)$ -th roots of unity, $P(\varepsilon, i) = P(\varepsilon \cdot i)$, there are numbers whose relative norm with respect to $P(\varepsilon)$ is exactly -1 .

The number $1 + i\varepsilon^\nu$ ($\nu = 1, 2, \dots, p-1$) has relative norm $(1 + i\varepsilon^\nu)(1 - i\varepsilon^\nu) = 1 + \varepsilon^{2\nu}$; consequently all numbers $1 + \varepsilon^\nu$ ($\nu = 1, 2, \dots, p-1$) are relative norms of numbers from $P(\varepsilon i)$ relative to the ground field $P(\varepsilon)$. Therefore the same holds for

$$\Pi = \prod_{\nu=0}^{r-1} (1 + \varepsilon^{2^\nu}) = (1 + \varepsilon)(1 + \varepsilon^2)(1 + \varepsilon^4) \cdots (1 + \varepsilon^{2^{r-1}}) = \sum_{\lambda=0}^{2^r-1} \varepsilon^\lambda.$$

If one adjoins to Π the term $\varepsilon^{2^r} = \varepsilon^{-1} = \varepsilon^{p-1}$ (by $(*)$), then $\Pi + \varepsilon^{p-1}$ becomes a sum of pieces $1 + \varepsilon + \varepsilon^2 + \cdots + \varepsilon^{p-1} = 0$, and consequently

$$\Pi = -\varepsilon^{-1}.$$

¹¹The characterization of the splitting fields can also be obtained easily with the help of factor systems.

Now ε too is the relative norm of a number from $P(\varepsilon i)$, namely of $\varepsilon^{(p+1)/2}$. Hence the analogous statement holds for

$$\Pi\varepsilon = -1.$$

Therefore -1 is representable in $P(\varepsilon)$ as a sum of two squares;¹² $P(\varepsilon)$ is a splitting field for the quaternion group.

Let K be the cyclic field of degree 2^n contained in $P(\varepsilon)$. We assert that K too is a splitting field.¹³ Let m be the index of the quaternion division ring with respect to K as ground field. Then $m = 1$ or $m = 2$. But since there is a splitting field $P(\varepsilon)$ of odd relative degree $(p-1)/2^n$ with respect to K , the second case is impossible; hence $m = 1$. This means precisely that K is a splitting field.

Thus there are, for infinitely many n , cyclic fields of degree 2^n which are minimal splitting fields.

33. Hypercomplex quantities and representation theory in arithmetic conception

Atti Congresso Bologna 2 (1928), pp. 71–73.

Introduction

The most important general theorems on hypercomplex systems go back to Molien (Math. Annalen vol. 41 and Dorpat Reports 1897). Shortly afterward, essentially independently of this, Frobenius developed the theory of hypercomplex systems and of their representations – in particular the representation theory of finite groups – in a unified way. The foundation is the concept, going back to Dedekind, of the group determinant, and more generally of the determinant of an arbitrary hypercomplex system. Frobenius shows that the different irreducible factors of this determinant (the coefficient domain is always the field of all complex numbers) correspond to the different irreducible representation classes, and that in this way all irreducible representation classes are exhausted. Such a representation class is completely characterized by its “character system”; in the case of finite groups this character system arises by decomposition of the determinant of the commutative hypercomplex system derived from the classes of conjugate elements of the group. This determinant decomposes into linear factors with the characters as coefficients – a direct generalization of the result first obtained by Dedekind, that the group determinant of a finite abelian group decomposes into linear factors whose coefficients are the various characters of the abelian group (correspondence with Frobenius).¹⁴

The totality of possible representations – perhaps only of the irreducible ones – of a given ring in a given, not necessarily commutative, division ring is under consideration.

The first problem is by far the simpler one; this is already shown by the fact that no special hypotheses of any kind are needed either on the ring to be represented or on the – in general non-commutative – representation division ring. The fact that the representation is by matrices of fixed degree supplies all the necessary finiteness hypotheses. The problem can be treated completely by means of the theory of groups with operators, and I shall therefore briefly discuss it.

A group $\mathfrak{M}$ is called a group with a domain of operators if, at the same time, there is given a system of symbols $\Theta, H, \dots$ – the operators – such that the combination $\Theta(a), H(a), \dots$ is uniquely defined for every a in $\mathfrak{M}$ and yields an element of $\mathfrak{M}$, and such that the distributive law is satisfied: $\Theta(ab) = \Theta(a)\Theta(b)$. Two groups with the same domain of operators are called operator-isomorphic if they are isomorphic in the usual sense and if, moreover, from the correspondence of a and $\bar{a}$ there follows also that of $\Theta(a)$ and $\Theta(\bar{a})$, of $H(a)$ and $H(\bar{a})$, and so on. If as subgroups one then admits only those which themselves admit the domain of operators (so that a group is called simple if it has no admissible proper normal divisor except the identity), then the Jordan-Hölder theorem on composition series remains valid in the form: if a group with operator domain has a composition series at all, then the system of factor groups is uniquely determined up to ordering in the sense of operator-isomorphism. Under the same hypothesis, there also holds the theorem on the decomposition, unique in the sense of operator-isomorphism, into directly indecomposable factors.

The connection with the representation problem is given by the fact that, among the groups with operators, one finds in particular ideals and modules, these being regarded as abelian groups with respect to addition,

¹²For primes of the form $8n \pm 3$ this argument is already found in E. Landau, Über die Darstellung definiter Funktionen durch Quadrate, Math. Ann. vol. 62, pp. 272–285.

¹³The choice of the cyclic splitting field of degree 2^n as a subfield of a cyclotomic field is modeled on Hasse’s example.

¹⁴What follows is a free elaboration, prepared by B. L. van der Waerden, of my lecture from the winter semester 1927/28. We undertook the preparation for print jointly. I am also indebted to B. L. van der Waerden for a number of critical remarks.

while the operators are given by multiplication with the ring elements. Every representation, however, is generated by a representation module, i.e. by a module with respect to the ring and the representation division ring. More precisely, every module class, i.e. every class of representation modules operator-isomorphic to one another, generates the same representation. Thus the question of uniqueness of the reduction of a representation is answered by the composition-series theorem and by the theorem on the direct product. Applied to the representation module, the composition series gives, for each matrix, a reduction of the form

$$\begin{pmatrix} B_1 & 0 & 0 \\ 0 & B_2 & 0 \\ A_{ik} & \cdots & B_r \end{pmatrix},$$

where the diagonal matrices corresponding to the factor groups generate an “irreducible” representation, i.e. one that itself no longer admits such a reduction.

Since the classes of these factor groups are uniquely determined, the same holds for the diagonal representations (uniquely in the sense of equivalence of representations). Analogously, the theorem on the direct product gives the uniqueness of the “decomposition” into “indecomposable” constituents:

$$\begin{pmatrix} C_1 & & & \\ & C_2 & & \\ & & \ddots & \\ & & & C_s \end{pmatrix},$$

where each C_i can then still be reduced uniquely according to the composition-series theorem.

I would like to sketch the second problem for the case of representations of hypercomplex systems, more generally of rings, which satisfy the “double-chain theorem”. The result is that every irreducible (simple) module class is an ideal class; hence the composition series from one-sided ideals already supply all irreducible representations through their factor groups. More precisely, the composition series in the residue-class ring modulo the radical suffice, the latter becoming a “completely reducible” ring. The problem is thereby reduced to the structural investigation of completely reducible rings; and thus it proves appropriate to take as starting point the representations in the automorphism division rings. This is to be understood as follows. All simple (right) ideals of a two-sided simple component of such a ring belong to the same class, and therefore have the same automorphism ring, which, because of the simplicity of the ideals, becomes a division ring; at the same time it is the automorphism ring of the simple left ideals. The simple ring itself becomes isomorphic to the system of all matrices over the automorphism division ring; and from this representation spring all representations with respect to a subfield, in particular with respect to the coefficient domain of the hypercomplex system, by replacing the elements of the automorphism division ring themselves by assigned matrices. Thus the Wedderburn structure theorem known in the case of hypercomplex systems is reinterpreted here through the concept of the automorphism division ring, a concept originating in general group theory, and is at the same time recognized as the source of the representation theory of hypercomplex systems. The new questions of the Galois theory of non-commutative division rings arise; they are closely connected with the question of “splitting fields”. At the same time, an outlook is obtained toward structure theorems for general rings that are not completely reducible, by means of the automorphism rings of the indecomposable ideals. Group theory, in its extension to the most general groups with operators, here again proves itself to be the ordering principle.